

## **Criterion B – Inquiring and Designing**

### **1 – Research Question (Strand i)**

A student notices that an ice cube left on a plate in a warm classroom disappears over time. She wants to investigate whether the temperature of the surroundings affects how quickly the ice melts.

**Outline a clear and testable research question for this investigation.**

**How does the temperature of the surroundings affect the time taken for an ice cube to melt completely?**

- Independent variable: Temperature of the surroundings
- Dependent variable: Time taken for the ice cube to melt

A good research question should clearly show the relationship between the independent and dependent variables

### **2 – Hypothesis (Strand ii)**

**A student investigates whether the temperature of water affects how quickly liquid water evaporates. She plans to place equal volumes of water in identical shallow dishes at different temperatures and measure how much water evaporates after 30 minutes.**

**Using the Particle Theory, outline a testable prediction (hypothesis) for this investigation.**

**Your hypothesis should:**

- **predict the relationship between the variables.**
- **explain your prediction using scientific reasoning.**

**If the temperature of the water increases, then the amount of water that evaporates in 30 minutes will increase.**

**Scientific reasoning:**

According to the Particle Theory, particles in warmer water have more kinetic energy and move faster. Therefore, more particles at the surface have enough energy to escape from the liquid and become water vapour.

**Therefore:**

Higher temperature → more kinetic energy → faster-moving particles → more evaporation

**3 – Variables and Data Collection (Strand iii)**

**A student investigates how the temperature of water affects the time taken for a fixed-size ice cube to melt completely.**

**She has:**

- equal-sized ice cubes
- beakers of water at 10°C, 20°C, 30°C and 40°C
- a stopwatch
- a thermometer

**Outline:**

**a) the independent variable-Temperature of the water (°C)**

b) the dependent variable **Time taken for the ice cube to melt completely (seconds)**

c) three controlled variables- **Any three**

**Size/mass of the ice cube**

- **Volume of water**
- **Size/type of beaker**
- **Starting temperature of the ice cubes**
- **Same method for deciding when the ice has completely melted**

d) describe how you will record the data during the investigation.

| <b>Water temperature (°C)</b> | <b>Trial 1 (s)</b> | <b>Trial 2 (s)</b> | <b>Trial 3 (s)</b> | <b>Mean melting time (s)</b> |
|-------------------------------|--------------------|--------------------|--------------------|------------------------------|
|-------------------------------|--------------------|--------------------|--------------------|------------------------------|

10

20

30

40

#### 4 – Designing an Investigation (Strand iv)

**Design a scientific investigation to answer the following research question:**

**Does the temperature of surrounding water affect the time taken for a fixed-size ice cube to melt completely?**

**Include:**

- **Aim**
- **Hypothesis (with scientific reasoning)**
- **Variables (independent, dependent and controlled)**
- **Materials**
- **Step-by-step method**
- **Safety precautions**
- **A suitable results table**

#### **Materials**

- 4 identical beakers
- Water
- Equal-sized ice cubes
- Thermometer
- Stopwatch
- Measuring cylinder

#### **Method**

IB MYP  
Home task- 1 (2026-27)

1. Measure an equal volume of water and pour it into each beaker.
2. Adjust the water temperatures to 10°C, 20°C, 30°C and 40°C.
3. Check each temperature using a thermometer.
4. Place an equal-sized ice cube into the first beaker.
5. Start the stopwatch immediately.
6. Stop the stopwatch when the ice cube has completely melted.
7. Record the melting time in seconds.
8. Repeat the procedure for the other temperatures.
9. Repeat each temperature three times.
10. Calculate the mean melting time for each temperature

### Safety precautions

- Handle glass beakers carefully.
- Wipe up spilled water immediately to prevent slipping.
- Handle warmer water carefully.

### Results table

| Water temperature (°C) | Trial 1 (s) | Trial 2 (s) | Trial 3 (s) | Mean melting time (s) |
|------------------------|-------------|-------------|-------------|-----------------------|
| 10                     |             |             |             |                       |
| 20                     |             |             |             |                       |
| 30                     |             |             |             |                       |

40

**Criterion C – Processing and Evaluating**

**5 – Recording Data (Strand i)**

The table below shows the mean melting time of an ice cube at different water temperatures.

| Water Temperature (°C) | Mean Melting Time (s) |
|------------------------|-----------------------|
| 10                     | 550                   |
| 20                     | 390                   |
| 30                     | 230                   |
| 40                     | 145                   |

Construct a suitable line graph using the data provided.

Remember to include:

- title
- labelled axes
- appropriate scale
- units

## Title: Effect of Water Temperature on the Melting Time of an Ice Cube

**X-axis: Water Temperature (°C)**

**Y-axis: Mean Melting Time (s)**

### 6 – Interpreting Results (Strands ii )

Using the data above,

Interpret the results and outline the relationship between water temperature and melting time.

Use the Particle Theory to explain the trend.

The results show that as the water temperature increases, the melting time decreases. For example, at 10°C, the ice cube takes 550 seconds to melt, whereas at 40°C, it takes only 145 seconds.

At higher temperatures, water particles have more kinetic energy and move faster.

They transfer thermal energy to the ice more quickly. Therefore, the ice melts faster and takes less time to melt completely.

So the relationship is:

Higher water temperature → lower melting time

### 7 – Evaluating the Hypothesis (Strand iii)

Before conducting the investigation, the student predicted:

"As the temperature of the surrounding water increases, the ice cube will melt faster because warmer water particles have more kinetic energy and transfer heat to the ice more quickly."

Using the results,

Discuss whether the evidence supports the hypothesis.

Support your answer using data from the investigation.

As the water temperature increased, the time taken for the ice cube to melt decreased. At **10°C**, the ice cube took **550 seconds** to melt, while at **40°C**, it took only **145 seconds**.

## 8 – Evaluating the Method (Strand iv)

The student used the following method:

- She placed one 20 g ice cube into a beaker of water at each of four temperatures (10°C, 20°C, 30°C and 40°C).
- She used a stopwatch and stopped it when no solid ice was visible.
- She completed only one trial at each temperature.
- She did not control the starting temperature of the ice cubes.
- She used a standard kitchen thermometer to measure water temperature.

Discuss the validity of this method by identifying its strengths and weaknesses.

Consider:

- fairness of the investigation
- reliability



- accuracy
- control of variables

**The method used one 20 g ice cube at each temperature, only one trial, did not control the starting temperature of the ice, and used a standard kitchen thermometer.**

## **Strengths**

**One strength is that the same mass of ice cube (20 g) was used at each temperature.**

**This makes the investigation fairer because the size/mass of the ice cube was controlled.**

**Another strength is that a stopwatch was used to measure melting time.**

## **Weaknesses**

### **1. Only one trial**

**Only one trial was completed at each temperature. This reduces the reliability of the results because unusual results cannot be identified and a mean cannot be calculated.**

### **2. Starting temperature of ice cubes**

**The starting temperature of the ice cubes was not controlled. Some ice cubes may therefore have been colder than others, which could affect their melting time and reduce the validity of the investigation.**

## 9 – Improvements and Extension (Strand iv)

Based on your evaluation of the investigation,

Describe an improvement that would increase the reliability and accuracy of the investigation.

Then describe one extension that could further develop the investigation.

### Improvement

**Repeat the investigation three times at each temperature and calculate the mean melting time.**

### Why?

**Repeating the experiment makes the results more reliable because unusual results can be identified and the mean reduces the effect of random errors.**

### Extension

**Investigate a wider range of water temperatures, such as 5°C, 15°C, 25°C, 35°C and 45°C, to see whether the same relationship continues.**

**Investigate how the size or mass of an ice cube affects its melting time while keeping the water temperature constant.**